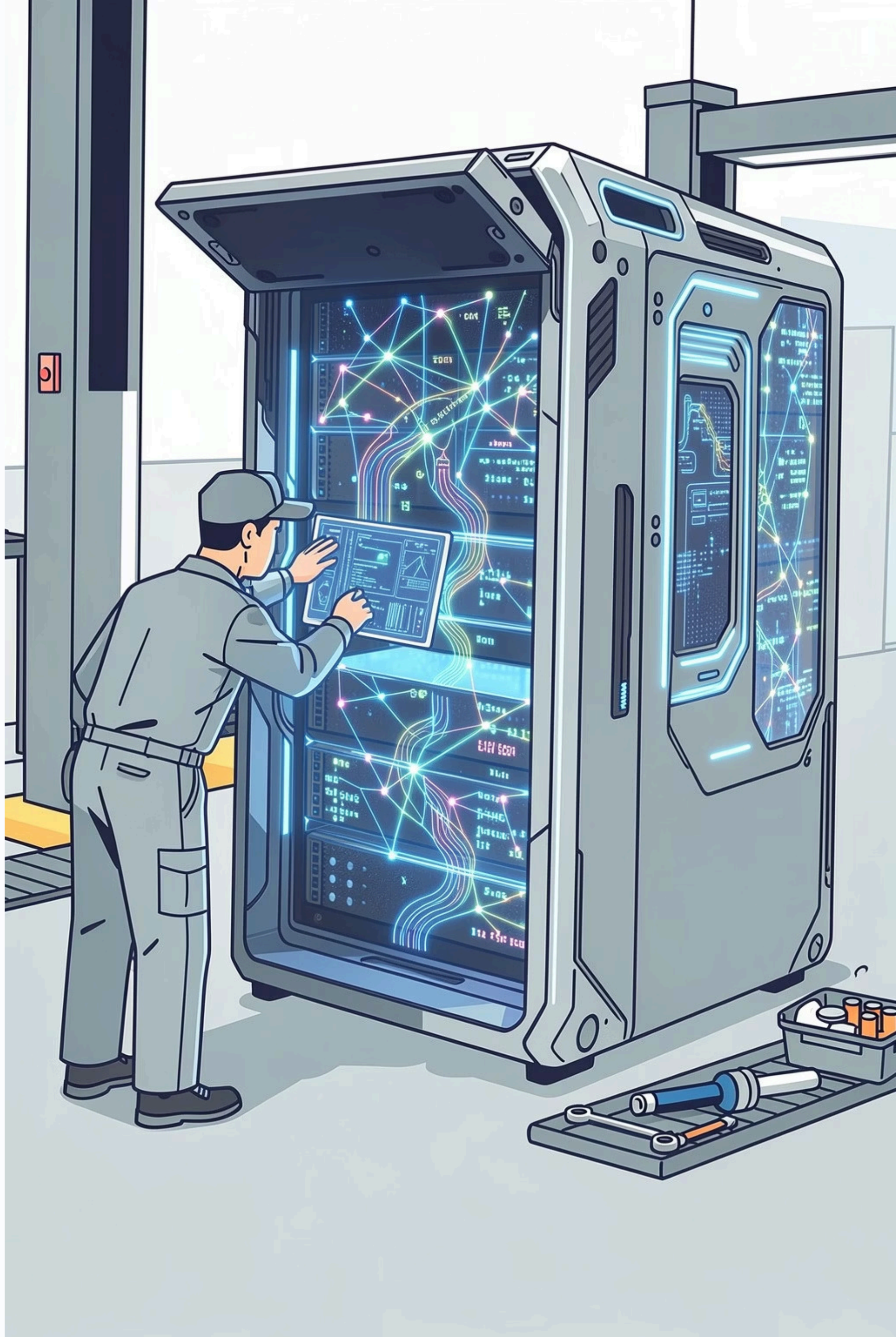
Session 8: Large Language Models (LLMs)

Last time you used ChatGPT and Gemini to create stuff. Today, we open the hood to see what's actually **inside** them. How do they sound so smart? How do they predict what to say? Let's find out.



Objectives

By the end of this session, you will be able to:

01

Define a Language Model

Explain what a Language Model is in your own words

02

See AI Autocomplete in Action

Understand how AI predicts the next word - like super-powered autocomplete

03

Understand Tokens

Know what tokens are - the building blocks of AI language

04

Why AI Sounds Human

Know why LLMs can sound so human (hint: it's NOT thinking!)

05

Compare ChatGPT vs Gemini

Put the two biggest LLMs side by side and spot the differences

Recap & Homework Share

Welcome back! Let's see your improved prompt.

- What did you change?

  Share how you modified your prompt from last session.

- Was the result better?

  Did the new prompt get you a better output from the AI?

- What did you learn?

  What did you discover about "talking" to AI?

Quick check from last session: What's Generative AI in your own words? Does AI actually *understand* what it creates? What's a prompt?

Warm-Up - Have You Noticed?

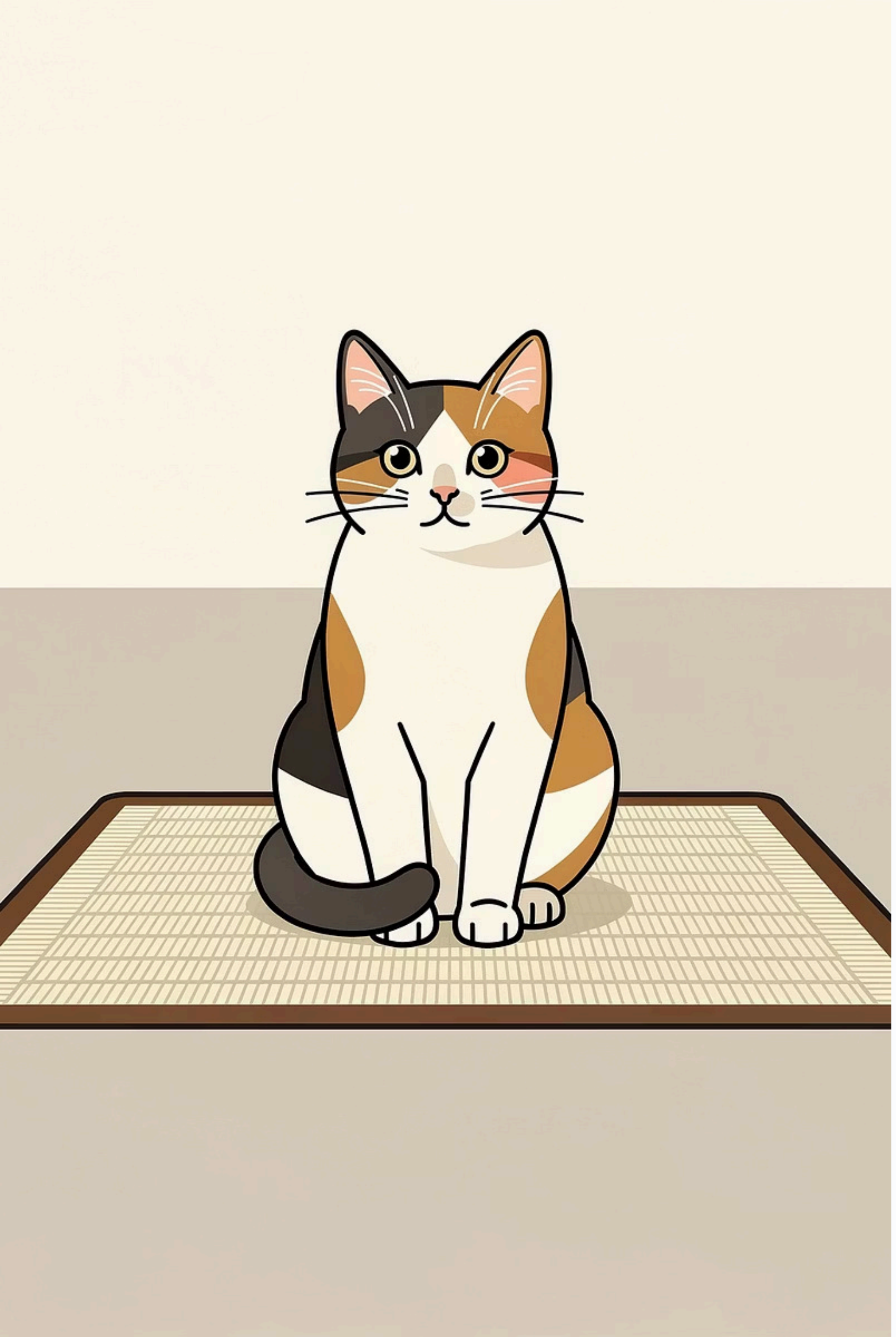
Think about your phone:

When you start typing a text message... your phone suggests the next word, right?

> 🗒️ You type "**See you**" → phone suggests "**later**" or "**tomorrow**"

🤔 **How does it know?**

Big question for today: How does ChatGPT do something similar - but for whole conversations?



What is a Language Model?

Language Model is an AI that predicts the next word.

Fill in the blank:

"The cat sat on the ____"

What word would YOU guess goes there?

Probably "**mat**" or "**chair**" or "**floor.**"

How did you know? Because you've heard sentences like that before. You learned the pattern.

Your Brain vs. a Language Model

Dimension	Your Brain	Language Model
How it learned	Hearing language	Billions of sentences
Prediction basis	Patterns AND meaning	Patterns only
Word meaning	Understands meaning	Doesn't understand - just matches patterns
Feelings & memories	Has both	No feelings, no memories
Awareness of topic	Knows what it's writing about	Doesn't know - just guesses really well

You think. AI matches patterns. Both can finish a sentence - but only YOU know what it means.

What Does "Large" Mean in LLM?

LLM = Large Language Model. The "L" for "Large" is real - these things are HUGE.

📚 Billions of Web Pages

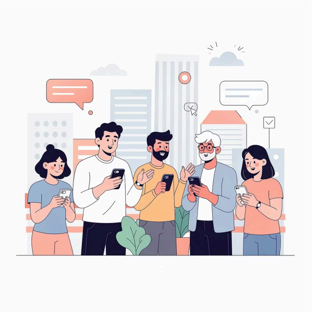
The entire public internet, scraped and processed

📖 Thousands of Books

Fiction, non-fiction, textbooks, and more

📰 Millions of Articles

News, research papers, blog posts



💬 Billions of Conversations

Forums, social media, Q&A sites

Imagine reading every book in your library. Then every book in every library. Then every webpage online. **That's what these AI systems have "read" before talking to you.**

How LLMs Are Trained

Here's how it works - like a HUGE game of fill-in-the-blank:

This process - repeated **billions of times** - is how AI gets great at guessing words. Every wrong guess makes it slightly better. Every right guess reinforces the pattern. After enough repetitions, the model becomes remarkably accurate at predicting language.

Mini Activity 1 - Predict the Next Word!

Your turn. What word comes next? Fill in the blank for each sentence:

Sentence	Your Guess
"Once upon a ____"	?
"I like to eat pizza with ____"	?
"To bake a cake, you need flour, eggs, and ____"	?
"The opposite of hot is ____"	?
"In the morning, I brush my ____"	?
"She opened her umbrella because it was ____"	?

This is exactly what a language model does - just with billions of examples to draw from!

How AI Sees It

Compare your guesses to what **most people** would guess. Notice the percentages - that's how often you'd hear that word in this context.

Sentence	Top Guess	Other Possibilities
Once upon a...	**time** (70%)	day, story
Pizza with...	**cheese** (60%)	pepperoni, friends
Flour, eggs, and...	**sugar** (50%)	butter, milk
Opposite of hot...	**cold** (95%)	freezing, warm
Brush my...	**teeth** (85%)	hair, face
Opened umbrella...	**raining** (80%)	rainy, wet

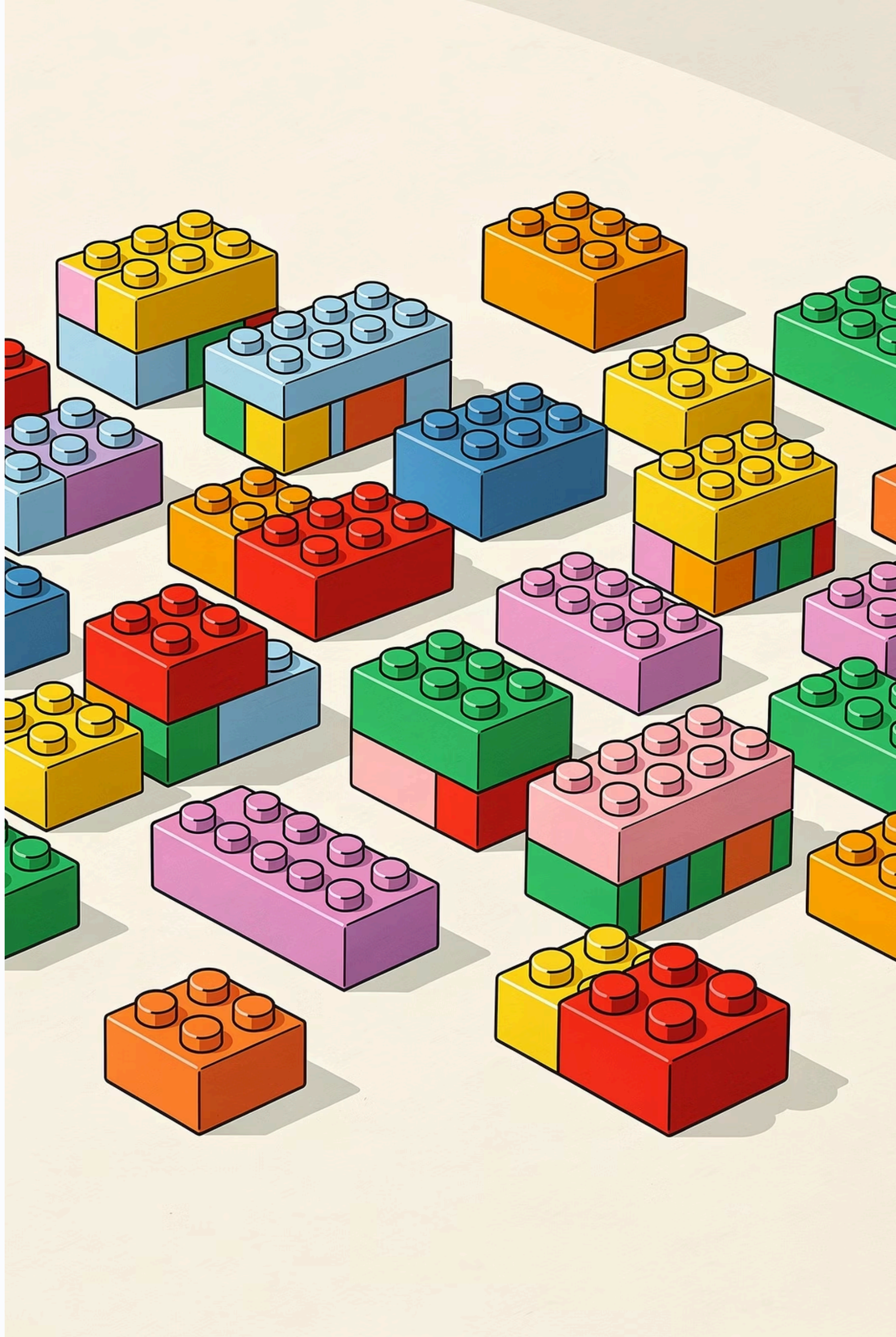
AI does the **EXACT same thing** - but with way more examples to compare against. Your brain is doing the same math, you just don't see the percentages!

One Tiny Thing - Tokens

AI doesn't actually predict whole words. It predicts pieces called **TOKENS**.

Word	Tokens (Pieces)
moonlight	[moon] + [light]
jumped	[jump] + [ed]
friendship	[friend] + [ship]
can't	[can] + ['t]

Tokens are like **LEGO** Blocks - Mix and match the blocks → you can build any word, even brand-new ones that have never existed before.



Why Does This Matter?

When you use **ChatGPT** or **Gemini**, you might see limits like **"1000 tokens per message."** That's because AI counts *pieces*, not whole words. Now you know exactly what they mean!

How AI Builds a Sentence

It's not magic. It's one word at a time. You ask: *"Tell me about dogs."*

Step 1

Most likely first word? → **"Dogs"**

Step 2

Next likely word? → **"are"**

Step 3

Next? → **"loyal"**

Step 4

Next? → **"animals"**

…and it keeps going, one token at a time. Like dropping breadcrumbs. **That's how ChatGPT writes whole paragraphs!**

Mini Activity 2 - YOU Be the Language Model!

You are now an LLM. Predict the next word!

Take turns with your mentor - one word at a time, building a sentence together.

Start with:

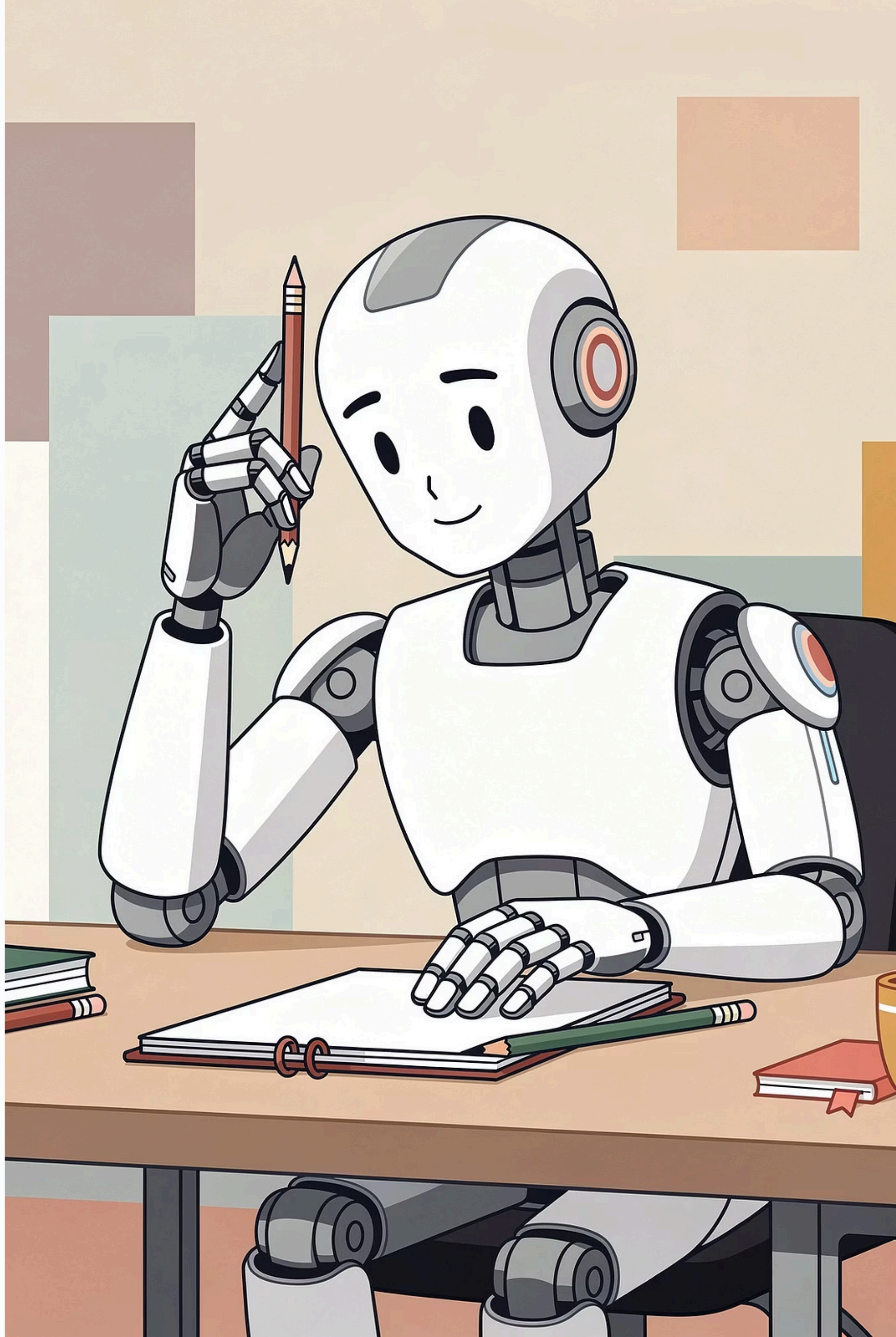
"The astronaut..."

How to play:

You add one word. Mentor adds the next word. You add one word. Mentor adds the next word. Keep going for 8–10 words.

Goal:

See what story you build together! Notice how the sentence takes on a life of its own.



Reflect - What Did You Notice?

Were you really "thinking" about the whole sentence?

Did you consciously plan every word before you started, or did it just... flow?

Or were you just picking what felt right next?

Each word led naturally to the next almost automatically, without a master plan.

Did the sentence end up making sense?

Surprisingly, it probably did - even though you never mapped it out from start to finish.

Here's the cool part: You probably **didn't plan the whole sentence**. You just picked one word at a time. **That's exactly what AI does. Just way, way faster.**

Why Does LLM Sound So Human?

Big question: If AI doesn't think... why does it feel like talking to a person?

01

It Learned From Humans

Billions of words written by humans - so it sounds exactly like us.

02

It Learned Conversations

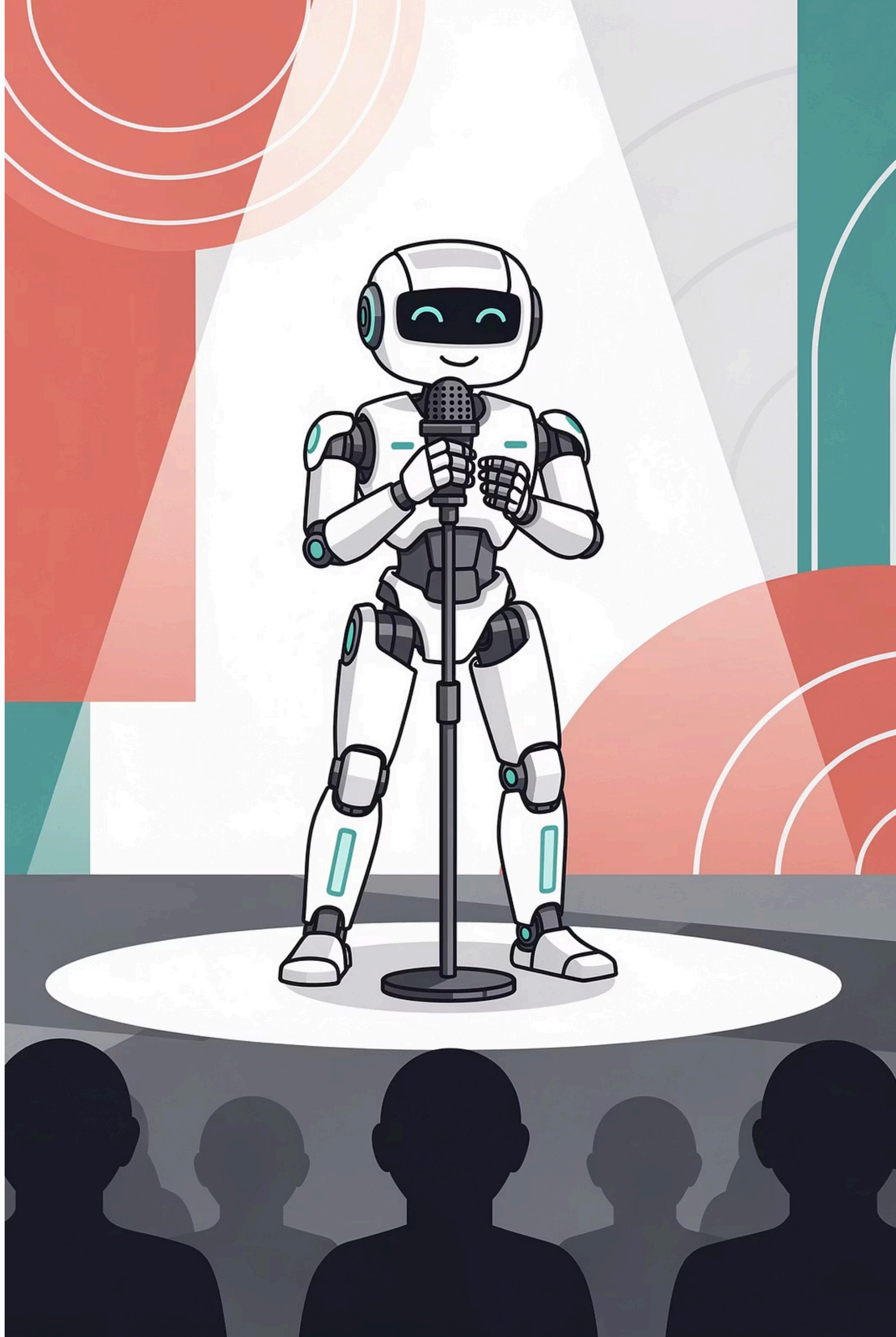
Millions of chats taught it how we greet, ask questions, joke, and respond.

03

It Learned Emotional Language

It knows how to *say* emotional things - even though it can't **feel** emotions.

⚠ **Bottom line:** It's a really, really good IMITATION. But there's no mind in there.



The Parrot Story

A parrot can say "Hello!" and "How are you?" and "Goodbye!" perfectly. **But does the parrot UNDERSTAND those words?** No. It just learned to repeat sounds that worked in certain situations.

LLMs are like SUPER advanced parrots. They've heard so many examples that they always say things that sound right. But they don't understand what they're saying — just like the parrot.

Parrot	LLM	Both
Repeats sounds that worked before	Repeats patterns that worked before	Sound convincing but don't truly understand

What LLMs Can Do

All from one skill: **predicting the next word.** LLMs are amazing at:

Answering Questions

"What's the capital of Brazil?"

Writing Stories & Poems

Creative writing of all kinds

Summarizing

Paste a long article, get a quick summary

Translating

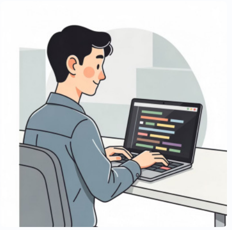
Between dozens of languages

Explaining Things Simply

"Explain gravity like I'm 10 years old"

Brainstorming Ideas

"Give me 10 names for my pet rabbit"



Helping with Code

Great if you're learning programming

What LLMs CAN'T Do (or Do Badly)

LLMs aren't perfect. They have real weaknesses - and knowing them makes you a smarter user.

No True Understanding

They pattern-match, not comprehend. There's no "thinking" happening.

Confidently Wrong

They can sound super sure when they're totally wrong! *(Next session: we call this "Hallucinations")*

Outdated Info

Even with web access, they sometimes mix old and new information about recent events.

They Forget

Long conversations confuse them - they lose track of earlier context.

Biases in Training Data

They can repeat biases from the data they were trained on.

No Real Feelings

They have no feelings - even when they sound caring or empathetic.

⚠ **Always double-check important stuff!**

Hands-On Activity - Compare ChatGPT vs Gemini!

ChatGPT and Gemini are BOTH Language Models. But they were trained differently different data, different settings, different "personalities."

Step 1: Open Both

Open **ChatGPT** (chat.openai.com) and **Gemini** (gemini.google.com) in two tabs

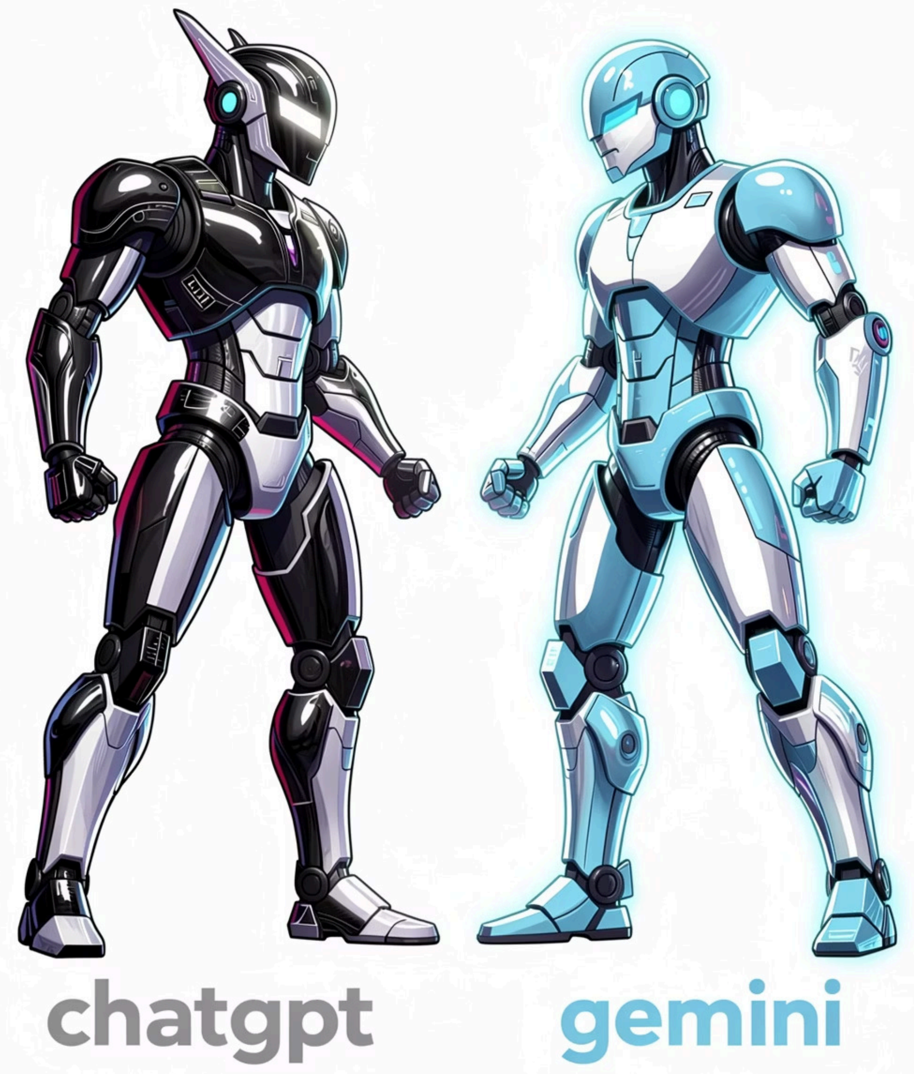
Step 2: Same Question

Ask the **exact same question** to both - word for word

Step 3: Compare

Look at what they say and notice the differences

What we're testing: Do they answer the same question the same way? If not - WHY?



Try These Prompts!

Ask **BOTH ChatGPT AND Gemini** the same things. Compare answers:

01

"Tell me a fun fact about space."

02

"Write a short, funny poem about a robot who loves pizza."

03

"Explain what AI is to a 5-year-old."

04

"Give me 3 ideas for a school project."

05

"Write a joke about computers."

Same or different?

Are the answers similar or totally different?

Which is better?

Funnier? More creative? Clearer?

Your pick?

Which one would YOU choose?

What Did You Notice?

Similar or Different?

Did ChatGPT and Gemini give similar answers? Or really different ones?

Personality Check

Did one feel more **creative**? More **friendly**? More **detailed**?

Right Tool for the Job

If you needed help with homework - which would you pick? If you wanted a fun story — which would you pick? Why?

Different LLMs have different strengths. The skill is picking the right one for the job.

Your Homework - LLM Detective 🕵️

Time to investigate ChatGPT and Gemini on your own. Choose **3 topics YOU find interesting** - anything goes!

Here are some ideas to get you started:

Your Favorite Sport

Football, basketball, swimming - whatever gets you excited!

A Video Game You Love

Minecraft, Roblox, Fortnite - any game you know inside and out.

An Animal You Like

Sharks, pandas, axolotls - pick your favorite creature!

A YouTuber You Follow

A creator you watch all the time - someone you know a lot about.

Ask Both ChatGPT and Gemini

Use the **same question** about each topic on both platforms.

Write Down Your Findings

Which gave the more **interesting** answer? Which gave the more **accurate** answer? Did either say anything wrong? Which one do YOU like better, and why?

Bonus Challenge

Ask ChatGPT or Gemini:

"Tell me something you DON'T know."

What does it say?

Most LLMs will say something like *"I don't know what's happening in the world right now"* or *"I don't have access to events after [date]."*

> ⚠ **That's actually important.** It means AI can sound super confident - but doesn't always know what it doesn't know.

Next Time: Session 9 - AI Hallucinations

You just learned that AI can be wrong. Next session, we go deep on **WHY** - and what to do about it.

What's an "AI Hallucination"?

Learn the term for when AI confidently makes things up

Why So Confident?

Why does AI say wrong things with total certainty?

Real Stories

When AI hallucinations got people in real trouble

Spot the Fake

How can YOU tell when AI is making stuff up?

Bring your LLM Detective notes! See you next session.